



Huawei to sue the FCC?

Huawei is reportedly preparing to sue the Federal Communications Commission over "unlawful" action. <https://dgit.in/dec19-49-1>



VR not a focus for Xbox

Microsoft say that Virtual Reality is not currently a focus for the Xbox team. <https://dgit.in/dec19-50>

- Windows:
 - Just Another Windows Script (JAWS)(<https://dgit.in/jaws>)
 - Watson (<https://dgit.in/watson>)
 - WindowsEnum (<https://dgit.in/winenum>)
 - Seatbelt (<https://dgit.in/seatbelt>)

ADVERSE EFFECTS OF PRIVILEGE ESCALATION ATTACKS AT THE ENTERPRISE LEVEL:

User access controls, permissions, privilege levels, etc play a huge role in a big enterprise where security is the utmost priority. Hence, in case of a breach, the main goal is to minimize the damage as much as possible. One policy to protect the systems is to implement the Principle Of Least Privilege, where only a bare minimum amount of permissions are given to a user which will allow them to do their work. So in case even if a user account is compromised, the damage will be only to the extent the permissions allow them. But privilege escalation attacks thwarts this technique completely and has an adverse effect if the hacker gets permissions not intended for anyone except the admins who manage the servers. Privilege escalation attacks are rarely the endpoints of a cyber-attack (it is different in case of HackTheBox, VulnHub, and other platforms, where root privileges are considered the final step) and after gaining the necessary permissions, the adversary usually has plans for other malicious activities like installing a backdoor, dumping crucial info, etc. Hence, in case of a breach, first responders should see to it that the attacker doesn't get far and trap him while the damage is minimal.

MITIGATION OF PRIVILEGE ESCALATION ATTACKS:

Now after having a look at privilege escalation attacks from the perspective of a hacker and considering the huge impact they have on enterprises, it time to put down the black hat and take up the white hat. Mitigation techniques overlap operating systems and exploit methodologies, so it's better to not do

any categorisation as everything applies to everything.

• Keeping applications and services patched and updated:

This is one of the crucial steps to prevent privilege escalation attacks. There are only a few cases where hackers use a zero-day vulnerability to leverage accesses because, in most scenarios, a pre-existing unpatched vulnerability is what gets used to gain entry. So keep your systems up-to-date and update them as soon as patches are released by the parent company of the vulnerable software. Especially, keep an eye out for security patches.

• Using safe practices while coding application:

Make sure to use functions that are not susceptible to common vulnerabilities like buffer overflow, stack smashing, etc. The safest way to ensure safety in this aspect is to use a memory-safe and type-safe language like java to write applications. Even if you are using C/C++, use the safe variants of vulnerable functions that analyse user-input before parsing them (for example, using `strncpy()` instead of `strcpy()`, which considers the buffer length before copying data into it).

• Input sanitization:

Wherever and whenever you are taking input from external sources, never trust it. Treat all input data as potentially malicious and perform necessary actions and checks before parsing it. This is true especially in case of attacks that use user input as attack vectors such as SQL injection, local file inclusion, etc which inevitably might lead to privilege escalation attacks.

• Implementing robust password policies:

This is the simplest way of them all, but the most difficult to enforce because human memory comes into play here. Passwords should be hard enough to be not guessed and invulnerable to brute-force attacks but easy enough to remember so that users don't keep Post-its notes of passwords stuck to their desks (this is just one of

the ways in which extreme password policies weaken the security instead of strengthening them). Also, make sure to change the default passwords of any hardware/software you use.

• Providing only minimal permissions:

As mentioned earlier, use the Principle Of Least Privilege while assigning permissions to ensure minimal damage. Create groups and users for specified tasks and localise them so as they can't access other areas. Same goes for file access permissions.

• Remove unused/unwanted functionalities:

Close all unused ports and services not actively used. Remove or restrict functionalities which might allow attackers to transfer data to and from the system. Remove accounts no more associated with active users and other dormant credentials. Restrict access to hardware devices like routers, printers and remove unused hardware from the network.

Following the above-mentioned practices will prevent/mitigate almost all the privilege escalation attacks. Stay alert and up-to-date with the latest vulnerabilities and hold a monthly or bi-monthly review to ensure all the ideal conditions are being satisfied. Make necessary changes whenever something out of order is discovered.

CONCLUSION:

Privilege escalation attacks, irrespective of operating systems and environment, are still a major issue in the cyber-security spectrum. And the problem can't be curbed unless the rules are strengthened, but then again compromises in performance and efficiency are unavoidable since humans are involved. A perfect balance, a sweet spot has to be found and the most important part is that system admins need to be aware and ditch the mentality "this cannot happen to my system". The enforcement will take time, but it is, indeed, possible to render the ever persistence privilege escalation attacks, void. 